Network Reference Model & TCP/IP Fundamentals

Unit-4

Protocol - Layered Approach

- **A protocol is a set of rules and conventions agreed upon and followed by the communicating entities for data communication.**
- **A protocol outlines the what, how and when of a communication.**

The three aspects of a protocol are −

- **Syntax − It defines the format of data that is to be sent or received.**
- **Semantics − It defines the meaning of each section of bits that are transferred.**
- **Timings − It defines the time at which data is transferred as well as the speed at which it is transferred.**

Protocol Hierarchies

Most networks are organized as a stack of layers, one on the top of another.

The number of layers and their names vary from network to network. Each layer has a specified function and adheres to specified protocols.

The below figure represents communication between Device A and Device B.

The data stream from one device to the other is not sent directly but has to pass through a number of layers.

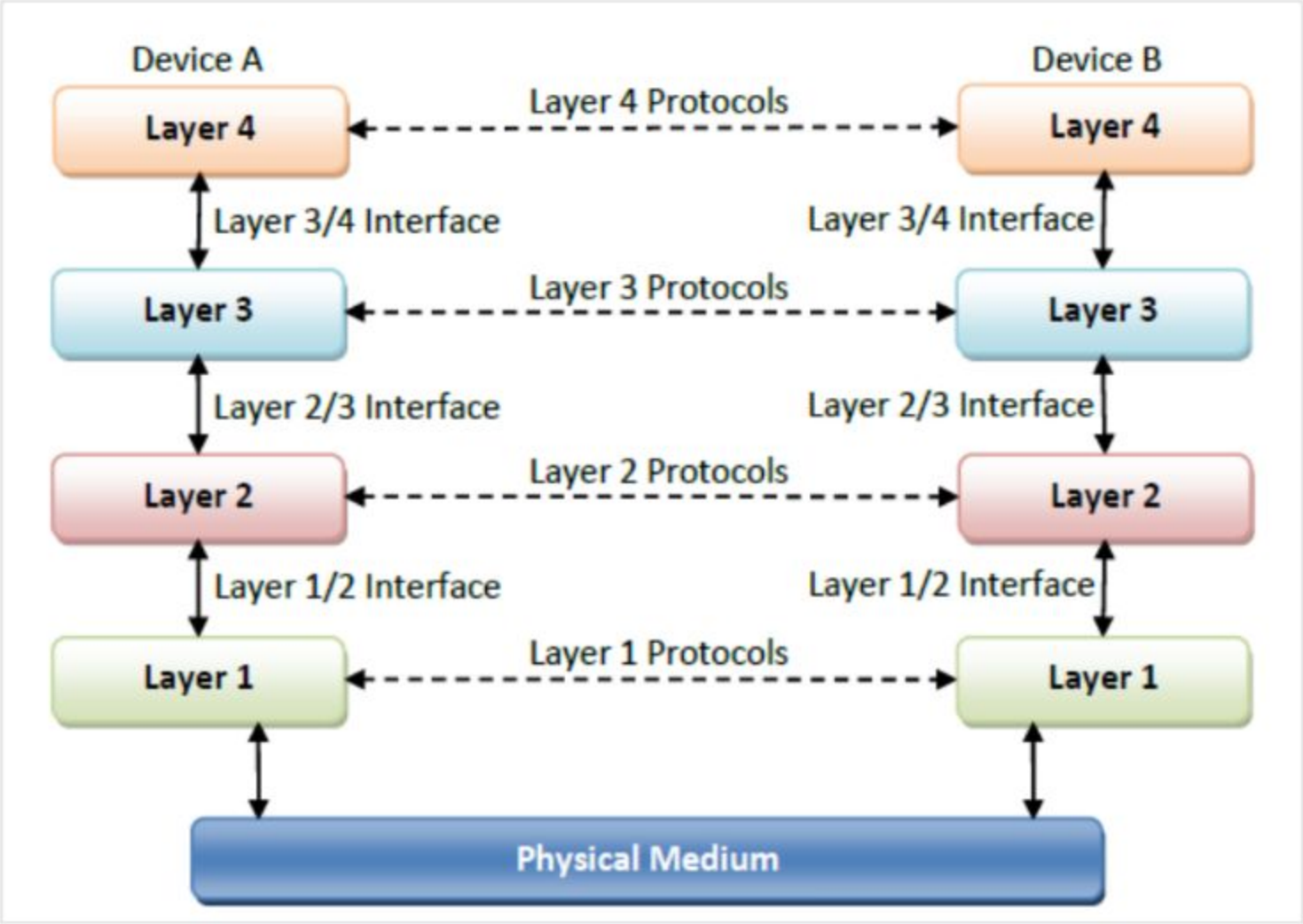
Protocol Hierarchies

The layers in the same levels are called peers and have a set of protocols for communication.

Between each adjacent layer is an interface that defines the services that are being offered by a lower layer to the next higher layer.

The dotted arrows depict virtual communication between peer layers, while the solid arrows represent

the physical communications between the adjacent layers.


Device A
Device B
Layer 4
Layer 4 Protocols
Layer 4
Layer 3/4 Interface
Layer 3/4 Interface
Layer 3
Layer 3 Protocols
Layer 3
Layer 2/3 Interface
Layer 2/3 Interface
Layer 2
Layer 2 Protocols
Layer 2
Layer 1/2 Interface
Layer 1/2 Interface
Layer 1
Layer 1 Protocols
Layer 1
Physical Medium

Interfaces between Layers

The passing of the data and network information down through the layers of the sending device and back up

through the layers of the receiving device is made possible by an interface between each pair of adjacent

layers.

Each interface defines what information and services a layer must provide for the layer above it. Well-defined

interfaces and layer functions provide modularity to a network

Services

Each layer at the sending site uses the services of the layer immediately below it.

The sender at the higher layer uses the services of the middle layer.

The middle layer uses the services of the lower layer.

The lower layer uses the services of the carrier.

Packet:

A packet is, generally speaking, the most basic unit that is transferred over a network. When

communicating over a network, packets are the envelopes that carry your data (in pieces)

from one end point to the other.

OSI reference Model: Layers & their Functions.

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards.

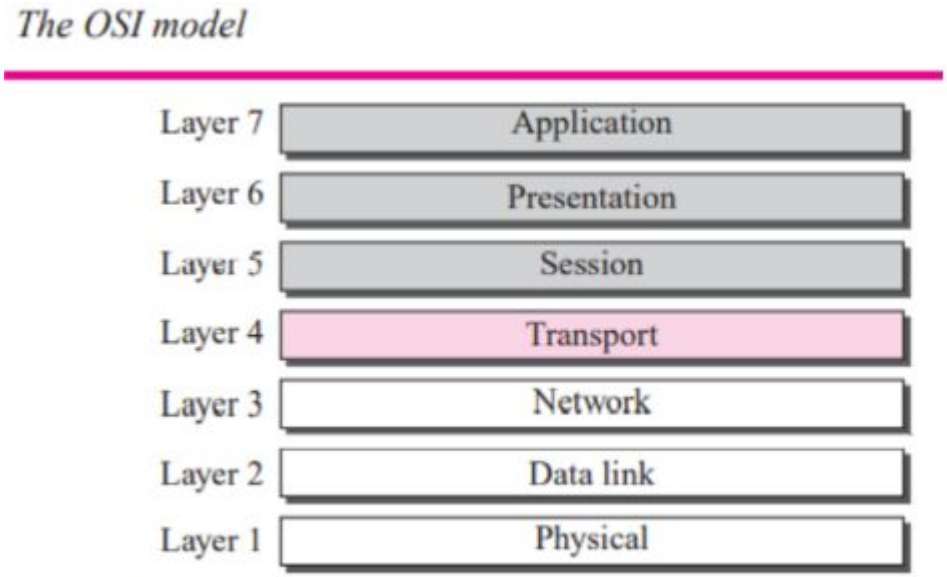
Almost three-fourths of countries in the world are represented in the ISO.

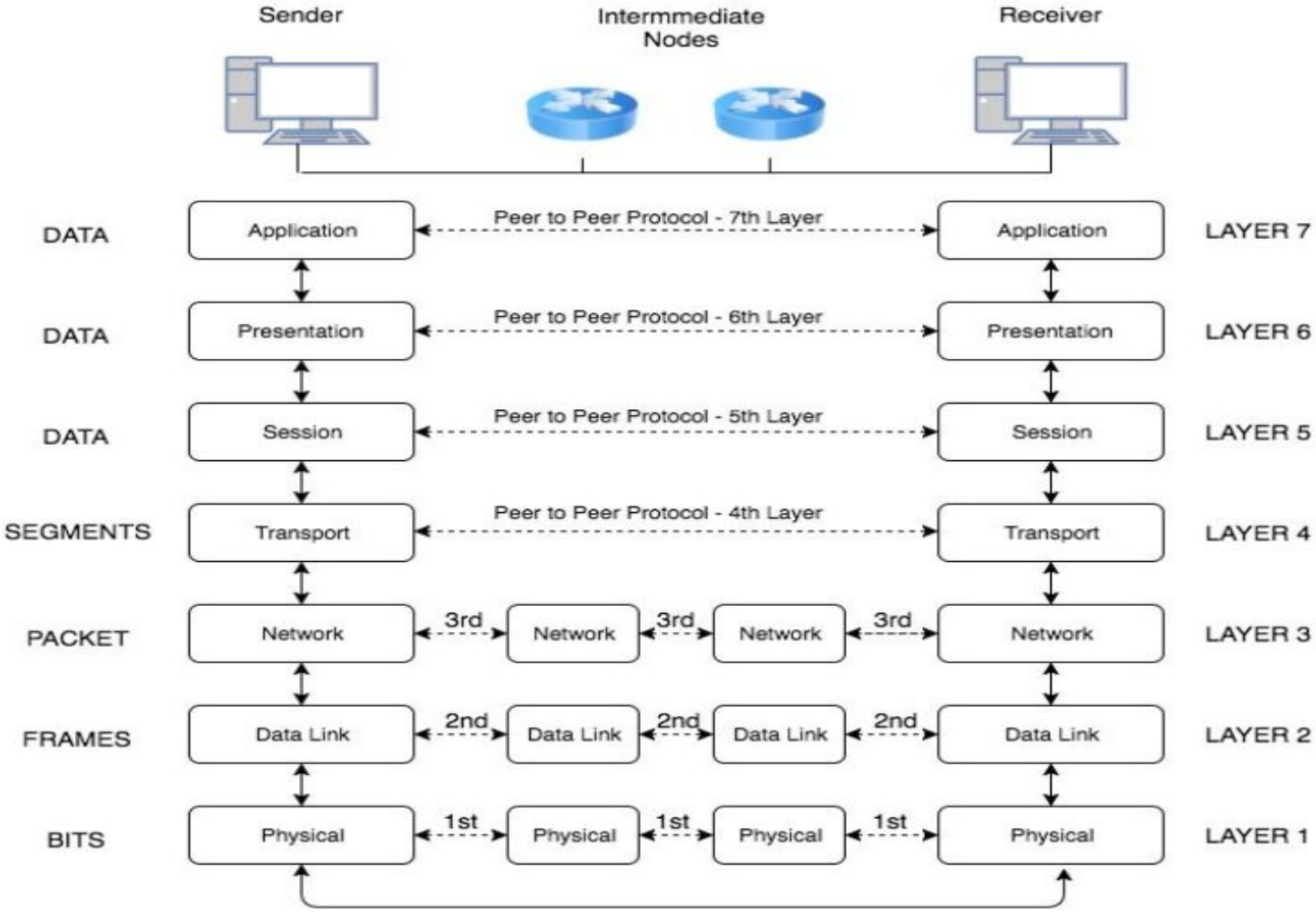
An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

The model is called the ISO OSI (Open Systems Interconnection) Reference Model because it deals with connecting open systems—that is, systems that are open for communication with other systems.

ISO is the organization; OSI is the model.

The OSI model has seven layers.




Sender
Intermmediate Nodes
Receiver
DATA
Application
Peer to Peer Protocol - 7th Layer
Application
LAYER 7
DATA
Presentation
Peer to Peer Protocol - 6th Layer
Presentation
LAYER 6
DATA
Session
Peer to Peer Protocol - 5th Layer
Session
LAYER 5
SEGMENTS
Transport
Peer to Peer Protocol - 4th Layer
Transport
LAYER 4
PACKET
Network
3rd
Network
3rd
Network
3rd
Network
LAYER 3
FRAMES
Data Link
2nd
Data Link
2nd
Data Link
2nd
Data Link
LAYER 2
BITS
Physical
1st
Physical
1st
Physical
1st
Physical
LAYER 1

Layered Architecture

The OSI model is composed of seven ordered layers: physical (layer 1),

data link (layer 2), network (layer 3), transport (layer 4), session (layer 5), presentation (layer 6), and application (layer 7).

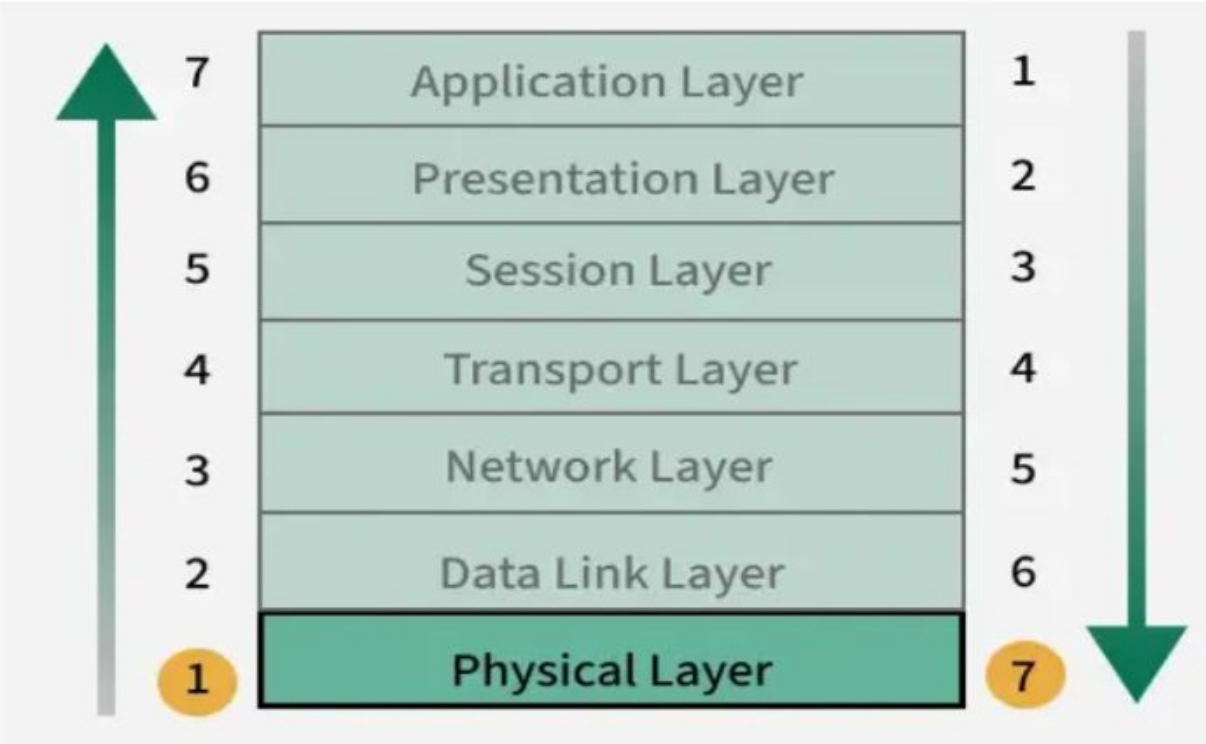
Figure shows the layers involved when a message is sent from device A to device B.

As the message travels from A to B, it may pass through many intermediate nodes.

These intermediate nodes usually involve only the first three layers of the OSI model.

Layers in the OSI Model

1. Physical Layer

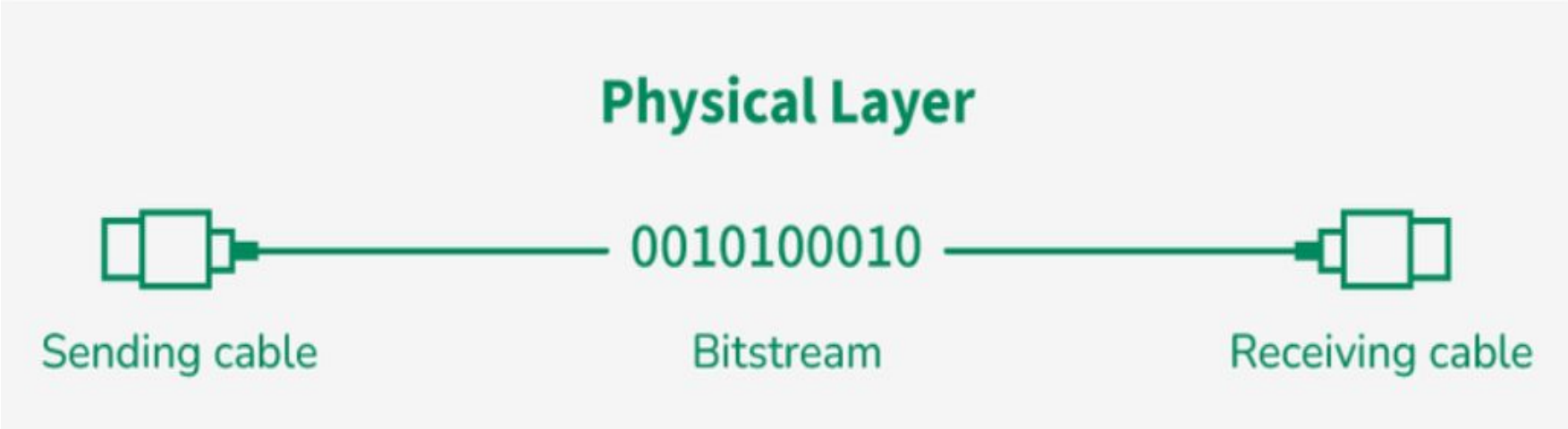


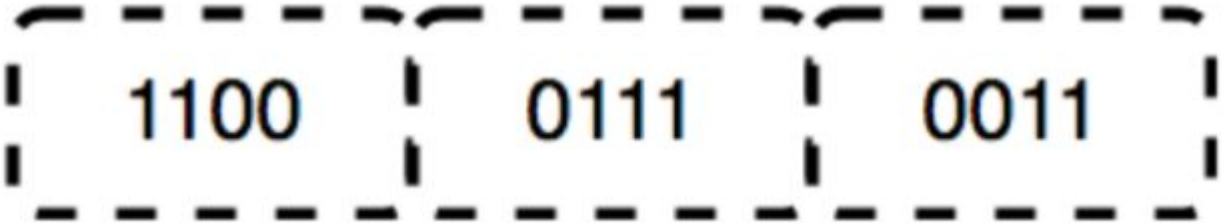
1. Physical Layer

- The physical Layer is the bottom-most layer in the **Open System Interconnection (OSI) Model** which is a physical and electrical representation of the system.
- It consists of various network components such as power plugs, connectors, receivers, cable types, etc.
- The physical layer sends data bits from one device(s) (like a computer) to another device(s).
- The physical Layer defines the types of encoding (that is how the 0's and 1's are encoded in a signal).
- It is responsible for the communication of the unstructured raw data streams over a physical medium

Functions Performed by Physical Layer

- The **Physical Layer** is responsible for sending raw data as bits over a physical medium.
- It converts data into signals that can travel through wires, fiber optics, or wireless channels (**encoding**) and turns these signals back into data at the receiver (**decoding**).
- It ensures signals are transmitted correctly and uses techniques like **modulation** to prepare the data for transmission and **demodulation** to retrieve it at the other end.
- This layer also decides how data flows (one-way, two-way alternately, or simultaneously) through **transmission modes** and controls the speed and timing of data transmission to keep everything running smoothly.


Physical Layer
0010100010
Sending cable
Bitstream
Receiving cable


1100
0111
0011

- **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at bit level.
- **Bit rate control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical topologies:** Physical layer specifies the way in which the different, devices/nodes are arranged in a network i.e. bus, star or mesh topology.
- **Transmission mode:** Physical layer also defines the way in which the data flows between the two connected devices.

The various transmission modes possible are: Simplex, half-duplex and full-duplex.

* Hub, Repeater, Modem, Cables are Physical Layer devices.

Protocols in Physical Layer

Typically, a combination of hardware and software programming makes up the physical layer. It consists of several protocols that control data transmissions on a network. The following are some examples of Layer 1 protocols:

- **Ethernet (IEEE 802.3)** – Widely used for wired networks.
- **Wi-Fi (IEEE 802.11)** – For wireless communication.
- **Bluetooth (IEEE 802.15.1)** – Short-range wireless communication.
- **USB (Universal Serial Bus)** – For connecting devices over short distances.

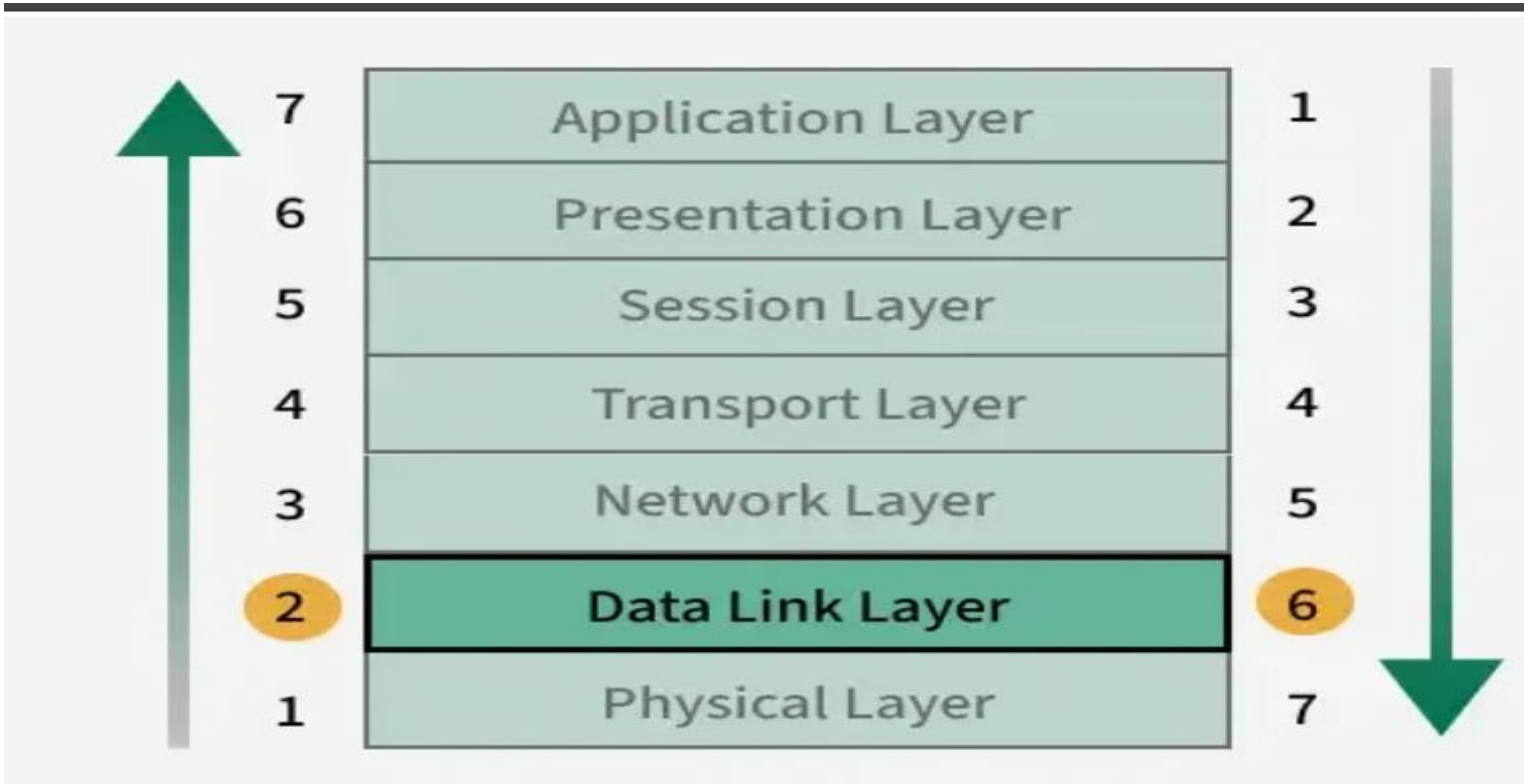
Advantages of the Physical Layer

- It ensures devices can transmit and receive raw data over physical mediums.
- It provides universal standards for cables, connectors, and signaling, ensuring compatibility.
- **Support for Various Media**: Works with wired (e.g., Ethernet) and wireless (e.g., Wi-Fi) technologies.

Limitations of the Physical Layer

- **No Error Handling**: Cannot detect or correct errors in data transmission.
- **Susceptible to Physical Damage**: Cables, connectors, and hardware failures can disrupt communication.
- **No Data Interpretation**: It only transmits bits and doesn’t understand or process the actual data.

Data Link Layer in OSI Model



Data Link Layer in OSI Model

The data link layer is the second layer from the bottom in the OSI (Open System Interconnection) network architecture model.

The data link layer is responsible for the node to node delivery of the message.

The main function of this layer is to make sure data transfer is error-free from one

node to another, over the physical layer.

When a packet arrives in a network, it is the responsibility of DLL to transmit it to

the Host using its MAC address.

Data Link Layer is divided into two sub layers :

1. Logical Link Control (LLC)

This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.

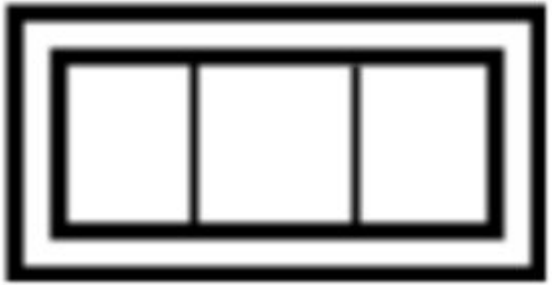
2. Media Access Control (MAC)

MAC sublayer manages the device’s interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

The packet received from Network layer is further divided into frames depending on the frame size of NIC (Network Interface Card).

DLL also encapsulates Sender and Receiver’s MAC address in the header.

The Receiver’s MAC address is obtained by placing an ARP(Address Resolution Protocol) request onto the wire asking “Who has that IP address?” and the destination host will reply with its MAC address.



The functions of the data Link layer are:

1. Framing: Framing is a function of the data link layer. It provides a way for a sender to

transmit a set of bits that are meaningful to the receiver. This can be accomplished by

attaching special bit patterns to the beginning and end of the frame.

2. Physical addressing: After creating frames, Data link layer adds physical addresses

(MAC address) of sender and/or receiver in the header of each frame.

3. Error control: Data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.

4. Flow Control: The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates that amount of data that can be sent before receiving acknowledgement.

5. Access control: When a single communication channel is shared by multiple devices, MAC sub-layer of data link layer helps to determine which device has control over the channel at a given time.* Packet in Data Link layer is referred as Frame.

** Data Link layer is handled by the NIC (Network Interface Card) and device drivers of host machines.

*** Switch & Bridge are Data Link Layer devices.

Protocols in Data link layer

- Ethernet, Point-to-Point Protocol (PPP), HDLC, MAC (Media Access Control).

  - Synchronous Data Link Protocol (SDLC)
  - High-Level Data Link Protocol (HDLC)
  - Serial Line Interface Protocol (SLIP)
  - Point to Point Protocol (PPP)
  - Link Access Procedure (LAP)
  - Link Control Protocol (LCP)
  - Network Control Protocol (NCP)

Limitations of Data Link Layer

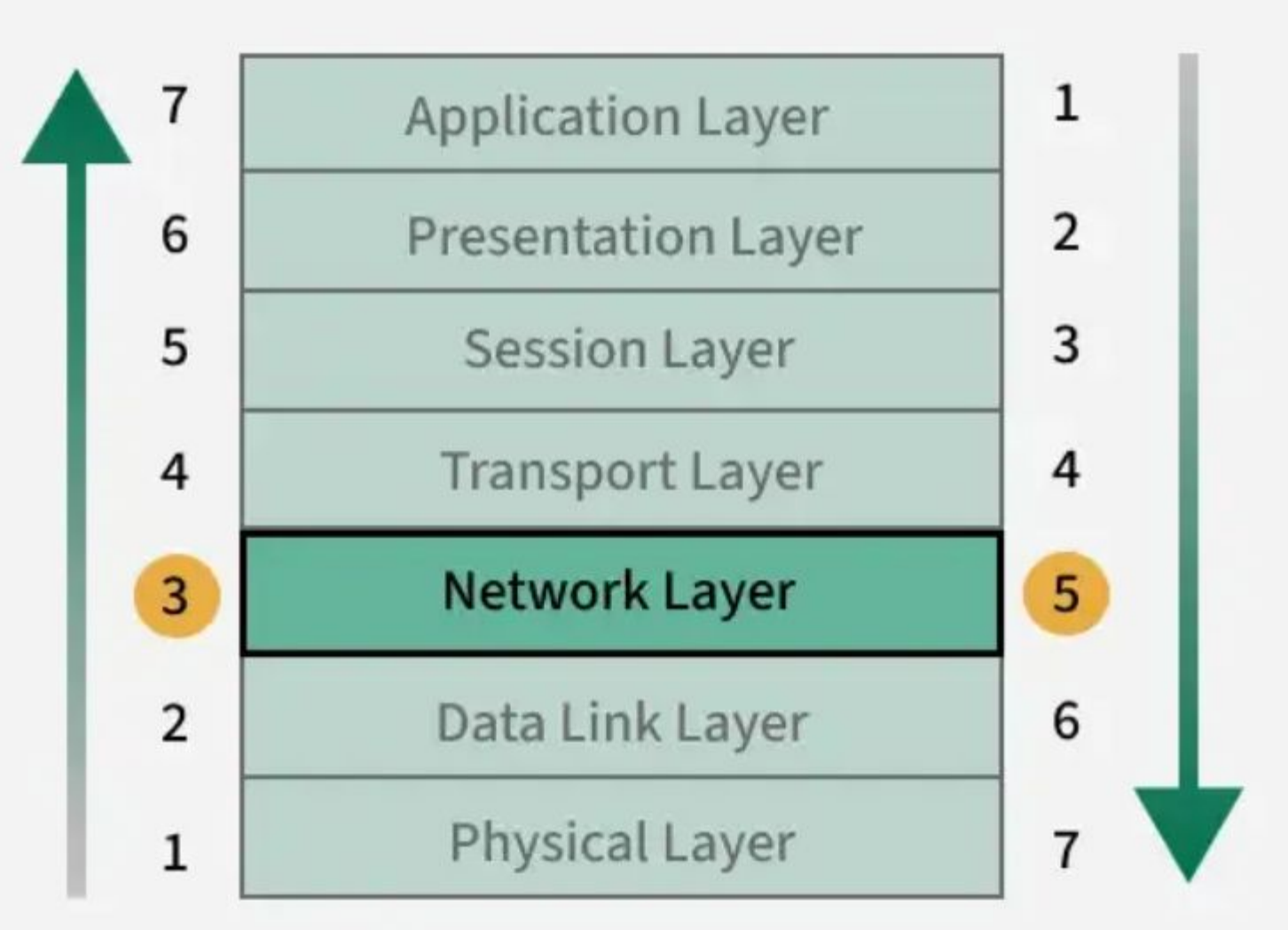
- **Limited Scope**: It operates only within a local network and cannot handle end-to-end communication across different networks.
- **Increased Overhead**: Adding headers, trailers, and redundant data (for error correction) increases the size of transmitted data.
- **Error Handling Dependency**: While it can detect and correct some errors, it relies on upper layers for handling more complex issues.
- **No Routing Capability**: The Data Link Layer cannot make routing decisions. It only ensures delivery within the same network segment.
- **Resource Usage**: Flow control and error correction mechanisms may consume extra processing power and memory

Applications of Data Link Layer

- **Local Area Networks (LANs)**: Enables reliable communication between devices within a local network using protocols like Ethernet (IEEE 802.3).
- **Wireless Networks (Wi-Fi)**: Manages communication between devices in wireless networks via protocols like IEEE 802.11 hence, handling media access and error control.
- **Switches and MAC Addressing**: Facilitates the operation of switches by using MAC addresses to forward data frames to the correct device within the network.
- **Point-to-Point Connections**: Used in protocols like PPP (Point-to-Point Protocol) to establish and manage direct communication between two nodes.

3. Network Layer (Layer 3):

- The Network Layer is the 5th Layer from the top and the 3rd layer from the Bottom of the OSI Model.
- It is one of the most important layers which plays a key role in data transmission.
- The main job of this layer is to maintain the quality of the data and pass and transmit it from its source to its destination. It also handles routing, which means that it chooses the best path to transmit the data from the source to its destination, not just transmitting the packet.
- There are several important protocols that work in this layer.


7
Application Layer
1
6
Presentation Layer
2
5
Session Layer
3
4
Transport Layer
4
3
Network Layer
5
2
Data Link Layer
6
1
Physical Layer
7

Data is transmitted in the form of packets via various logical network pathways between various devices.

It offers routes for data packet transfers across the network.

The network layer is also responsible for organizing and controlling the available paths for data transfer.

Functions of Network Layer

1. **Assigning Logical Address**: It provides unique IP addresses to devices for identification and communication across networks.
2. **Packetizing**: It encapsulates data into packets for efficient transmission.
3. **Host-to-Host Delivery**: It ensures data is delivered from the sender to the intended receiver across networks.
4. **Forwarding**: It is the process of moving packets from the input to the appropriate output interface in a router, based on the destination address

Functions of Network Layer

5. Fragmentation and Reassembly: It splits large packets into smaller fragments for transmission and reassembles them at the destination.

6.Logical Subnetting: It divides larger networks into smaller subnetworks for better management and routing efficiency.

7.Network Address Translation (NAT): Maps private IP addresses to a public IP for internet access, conserving IPs and adding security.

8.Routing: It determines the best path for packets to travel to their destination across multiple networks.

Working of Network Layer

- Every device gets a unique address (IP address) to identify it on the network.
- Data is packaged into small packets, with labels showing where it’s coming from and where it’s going.
- Routers figure out the best path to send the packets to their destination.

Working of Network Layer

- Packets travel step by step through different routers until they reach the right device.
- If a packet is too big, it gets broken into smaller pieces to fit through the network.
- At the destination, the pieces are put back together into the original data.
- If something goes wrong, like the destination can’t be reached, an error message is sent back.

Protocols Used at Network Layer

1. **IP (Internet Protocol)**
2. **ICMP (Internet Control Message Protocol)**
3. **ARP (Address Resolution Protocol)**
4. **RARP (Reverse Address Resolution Protocol)**
5. **NAT (Network Address Translation)**
6. **Routing Protocols**:
   - **RIP (Routing Information Protocol)**
   - **OSPF (Open Shortest Path First)**
   - **BGP (Border Gateway Protocol)**
7. **IPSec (Internet Protocol Security)**
8. **MPLS (Multiprotocol Label Switching)**

Advantages of Network Layer

- **Using the network layer in the OSI paradigm offers a multitude of advantages. Let's delve into some of these benefits:**
- **The network layer takes the data and breaks it down into packets, which makes transmitting the data over the network easier. This process also eliminates any weak points in the transmission, ensuring that the packet successfully reaches its intended destination.**
- **Router is the important component of the network layer . Its role is to reduce network congestion by facilitating collisions and broadcasting the domains within the network layer.**
- **Used to send data packets across the network nodes, the forwarding method is various.**

Limitations of Network Layer

- There is no flow control mechanism provided by the network layer design.
- There may be times when there are too many datagrams in transit over the network, causing congestion. This could put further strain on the network routers. In some circumstances, the router may lose some data packets if there are too many datagrams. Important data may be lost in the process of transmission as a result of this.
- Indirect control cannot be implemented at the network layer since the data packets are broken up before being sent. Additionally, this layer lacks effective error control systems.

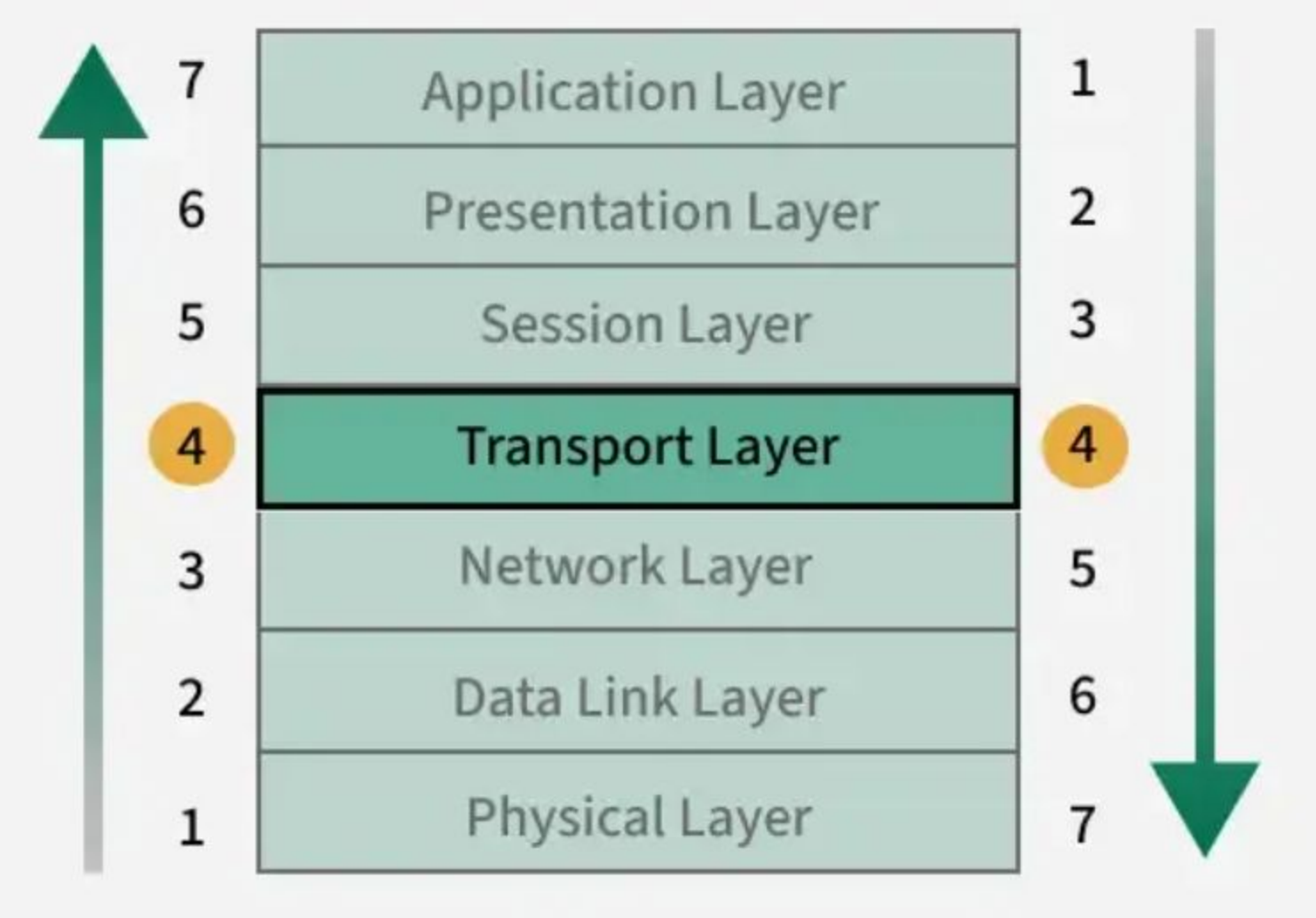
Transport Layer

Transport layer provides services to application layer and takes services from network layer.

The data in the transport layer is referred to as Segments.

It is responsible for the End to End Delivery of the complete message.

The transport layer also provides the acknowledgement of the successful data transmission and re-transmits the data if an error is found.


7
Application Layer
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Transport Layer
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Network Layer
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Data Link Layer
6
1
Physical Layer
7

At sender’s side:

o Transport layer receives the formatted data from the upper layers, performs Segmentation and also implements Flow & Error control to ensure proper data transmission.

o It also adds Source and Destination port number in its header and forwards the segmented data to the Network Layer.

At receiver’s side:

o Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.

The functions of the transport layer are:

1. Segmentation and Reassembly: This layer accepts the message from the (session) layer, breaks the message into smaller units. Each of the segment produced has a header associated with it. The transport layer at the destination station reassembles the message.

2. Service Point Addressing: In order to deliver the message to correct process, transport layer header includes a type of address called service point address or port address. Thus by specifying this address, transport layer makes sure that the message is delivered to the correct process.

The services provided by the transport layer:

1. Connection Oriented Service:

It is a three-phase process which include

– Connection Establishment

– Data Transfer

– Termination / disconnection

In this type of transmission, the receiving device sends an acknowledgement, back to the source after a packet or group of packet is received. This type of transmission is reliable and secure.

2. Connectionless service:

It is a one-phase process and includes Data Transfer.

In this type of transmission, the receiver does not acknowledge receipt of a packet.

This approach allows for much faster communication between devices.

Connection-oriented service is more reliable than connectionless Service.

-Data in the Transport Layer is called as Segments.

-Transport layer is operated by the Operating System.

It is a part of the OS and communicates with the Application Layer by making system calls.

Transport Layer is called as Heart of OSI model.

Transport Layer Protocols

Transport Layer Protocol uses different protocol for the better communication between two ends uses of protocol may differ from specifications. Below mention are some protocols used in Transport Layer

1. Transmission Control Protocol(TCP)

- TCP is connection-oriented Protocol.
- TCP is reliable protocol.
- As TCP is connection-oriented protocol, so first the connection is established between two ends and then data is transferred and then the connection is terminated after all data being sent.

2. User Datagram Protocol (UDP)

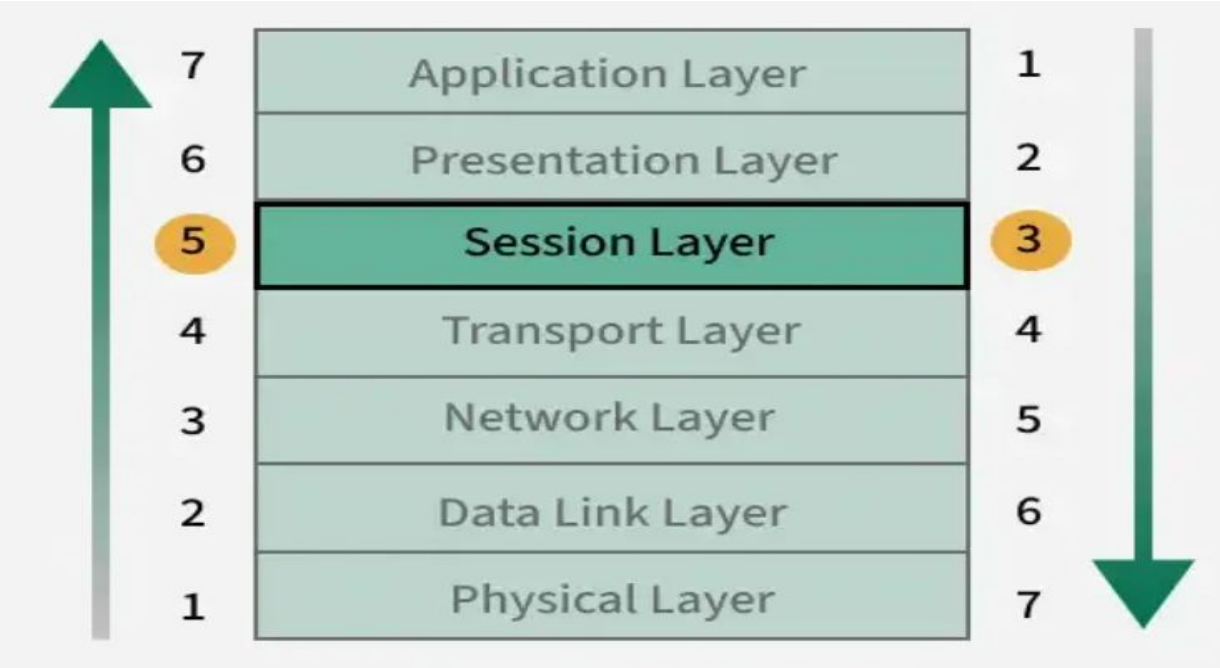
- UDP is not reliable protocol
- The protocol UDP is connectionless.
- When speed and size are more important than security and dependability, this kind of protocol is employed.
- The data from the higher layer is supplemented with transport-level addresses, checksum error control, and length information by UDP, an end-to-end transport level protocol.
- A user datagram is the packet that the UDP protocol generates.

3. **Stream Control Transmission Protocol (SCTP)**

- Many Internet applications use SCTP to perform transport layer duties, similar to User Datagram Protocol (UDP) and Transmission Control Protocol (TCP).
- On top of a connectionless packet network like IP, SCTP is a dependable transport protocol that facilitates data transfer over the network in scenarios involving one or more IP addresses

Session Layer

This layer is responsible for establishment of connection, maintenance of sessions,authentication and also ensures security.



Functions of Session Layer

1. **Session Establishment** : It establishes and manages sessions between communicating parties that cab be connection-oriented or connectionless. It also maps sessions to transport connections.
2. **Communication Synchronization** : It ensures reliable connectivity and recovery by using **synchronization bits** and **checkpoints** in data stream.
3. **Activity Management** : It allow the user to divide data into logical units called **activities**. An activity can be processed on its own and each activity is independent of activities that come before and after it.

Functions of Session Layer

4. Dialog Management : It refers to deciding whose turn it is to talk. Some applications uses a **token mechanism** for half-duplex mode, where only one party holds the token to transmit data while other supports full-duplex mode for simultaneous data transmission.

5. Data Transfer : It manages data exchange between systems.

6. Resynchronization : In this, all the tokens are restored to the positions that were set during synchronization. The various options for resynchronization includes set, abandon and restart.

Working of Session Layer

- The Session Layer manages communication sessions between applications over a network.
- It establishes connections, negotiating session parameters like authentication and communication direction (full-duplex or half-duplex).
- It oversees data exchange by using tokens to manage transmission rights and prevent collisions.

- Synchronization techniques are implemented, inserting checkpoints for recovery in case of disruptions.
- It ensures orderly communication, reducing message loss, duplication, or errors caused by overlapping communication.
- The Session Layer gracefully terminates the session, ensuring all data is exchanged and both sides agree to close

Session Layer Protocols

NetBIOS (Network Basic Input/Output System): Often used in local area networks (LANs), NetBIOS allows applications to communicate over a network by providing services such as naming, session management, and data transfer.

RPC (Remote Procedure Call): RPC allows programs to execute procedures on remote systems, giving the illusion of local execution.

PPTP (Point-to-Point Tunneling Protocol): Used for creating Virtual Private Network (VPN) connections, facilitating secure data transmission over the internet.

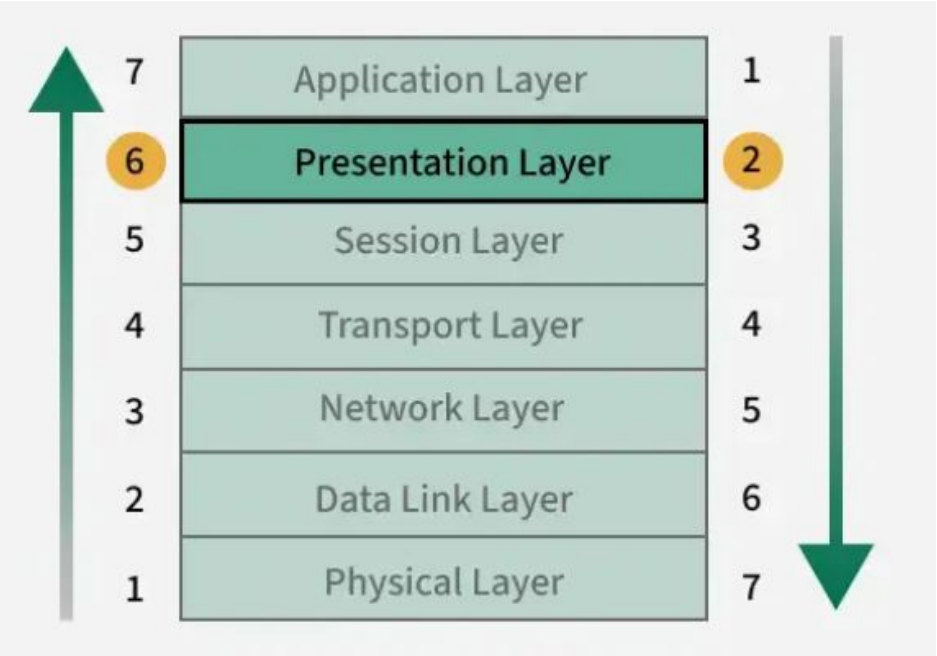
SMB (Server Message Block): A protocol that allows applications to read and write to files and request services from server programs in a network.

SQL (Structured Query Language) protocols: These manage and facilitate communication between databases and users or applications.

Presentation Layer

The Presentation Layer is the 6th layer of the OSI (Open Systems Interconnection) model. It sits between the Session Layer (Layer 5) and the Application Layer (Layer 7).

Its main responsibility is to translate, encrypt, and compress data between the application and transport layers.



Functions of the **Presentation Layer**

Data Translation:

- It ensures that the data sent from the application layer is in a format that can be understood by the receiving system.

- Common tasks include converting character sets (e.g., from EBCDIC to ASCII) or different data formats (e.g., JPEG to PNG).

Data Encryption

- The presentation layer can handle encryption and decryption of data to ensure secure communication. For example, **SSL/TLS** operates at this layer to secure data transmission over networks.

Functions of the **Presentation Layer**

Data Compression:

- It can also compress data to reduce the size of the data being transferred, which helps optimize network bandwidth and improve transmission speeds.
- Common compression algorithms include **ZIP** or **JPEG**.

Data Representation:

- It determines the way the data is represented. For instance, it can define the structure of data in terms of how numbers, characters, or even objects are encoded.

Services Provided by Presentation Layer

- Presentation layer could apply certain sophisticated compression techniques, so fewer bytes of data are required to represent the information when it is sent over the network.
- If two or more devices are communicating over an encrypted connection, then this presentation layer is responsible for adding encryption on the sender’s end as well as the decoding the encryption on the receiver’s end so that it can represent the application layer with unencrypted, readable data.

Services Provided by Presentation Layer

- This layer formats and encrypts data to be sent over a network, providing freedom from compatibility problems.
- This presentation layer is also responsible for compressing data it receives from the application layer before delivering it to the session layer (which is the 5th layer in the OSI model) and thus improves the speed as well as the efficiency of communication by minimizing the amount of the data to be transferred.

Protocols that operate at the Presentation Layer:

SSL/TLS (for encryption)

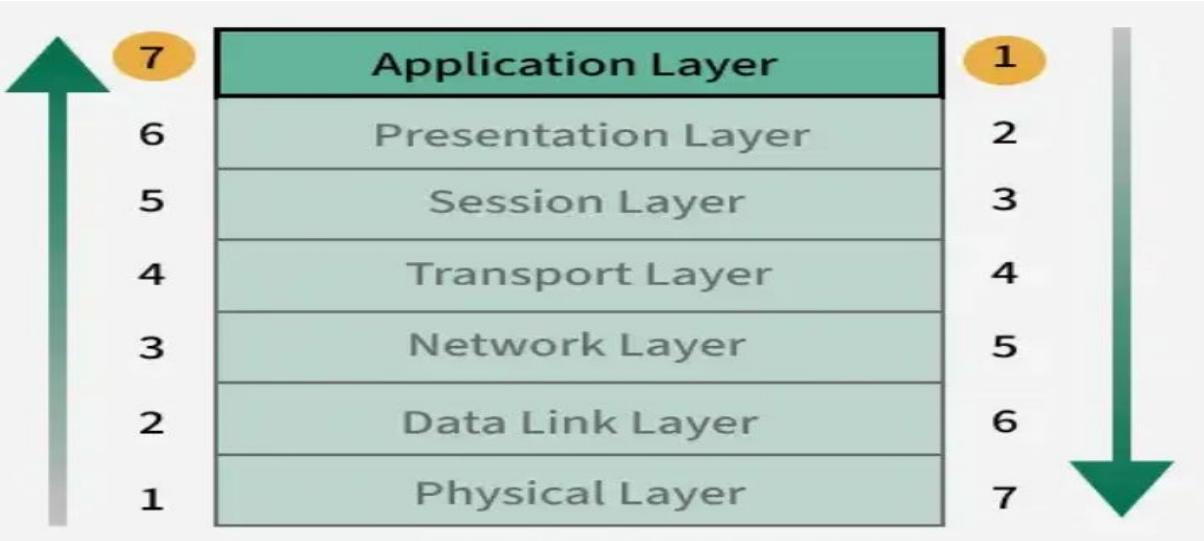
MIME (Multipurpose Internet Mail Extensions for data encoding)

JPEG, **GIF**, **PNG** (for image encoding)

Lightweight Presentation Protocol (LPP) Lightweight Presentation Protocol is that protocol which is used to provide ISO presentation services on the top of TCP/IP based protocol stacks.

Application Layer

The Application Layer of OSI (Open System Interconnection) model, is the top layer in this model and takes care of network communication. The application layer provides the functionality to send and receive data from users. It acts as the interface between the user and the application. The application provides services like file transmission, mail service, and many more.



Functions of Application Layer

1. Data Representation
2. Network Service Access
3. Application Protocols
4. Session Management

1. Data Representation

It ensures that data is in a readable format for both the sender and receiver, including data translation, compression, and encryption.

2.Network Service Access

It provides applications with access to network services, enabling communication over a network.

3.Application Protocols

It supports protocols like HTTP, FTP, SMTP, and DNS that enable communication between applications.

FTP

HTTP/S

SMTP

Telnet

4.Session Management

It manage establishment, maintenance, and termination of communication sessions between applications.

Working of Application Layer

- At first, client sends a command to server and when server receives that command, it allocates port number to client.
- Thereafter, the client sends an initiation connection request to server and when server receives request, it gives acknowledgement (ACK) to client through client has successfully established a connection with the server.
- Therefore, now client has access to server through which it may either ask server to send any types of files or other documents or it may upload some files or documents on server itself.

Services Provided by Application Layer Protocols

- The Application Layer protocol defines process for both parties which are involved in communication.
- These protocols define the type of message being sent or received from any side (either source host or destination host).
- These protocols also define basic syntax of the message being forwarded or retrieved.
- These protocols define the way to send a message and the expected response.
- These protocols also define interaction with the next level.

Protocols of the Application Layer

HTTP (HyperText Transfer Protocol) – Used for web browsing.

FTP (File Transfer Protocol) – Used for transferring files.

SMTP (Simple Mail Transfer Protocol) – Used for sending emails.

DNS (Domain Name System) – Resolves domain names to IP addresses.

IMAP (Internet Message Access Protocol) – Used for retrieving emails from a server.

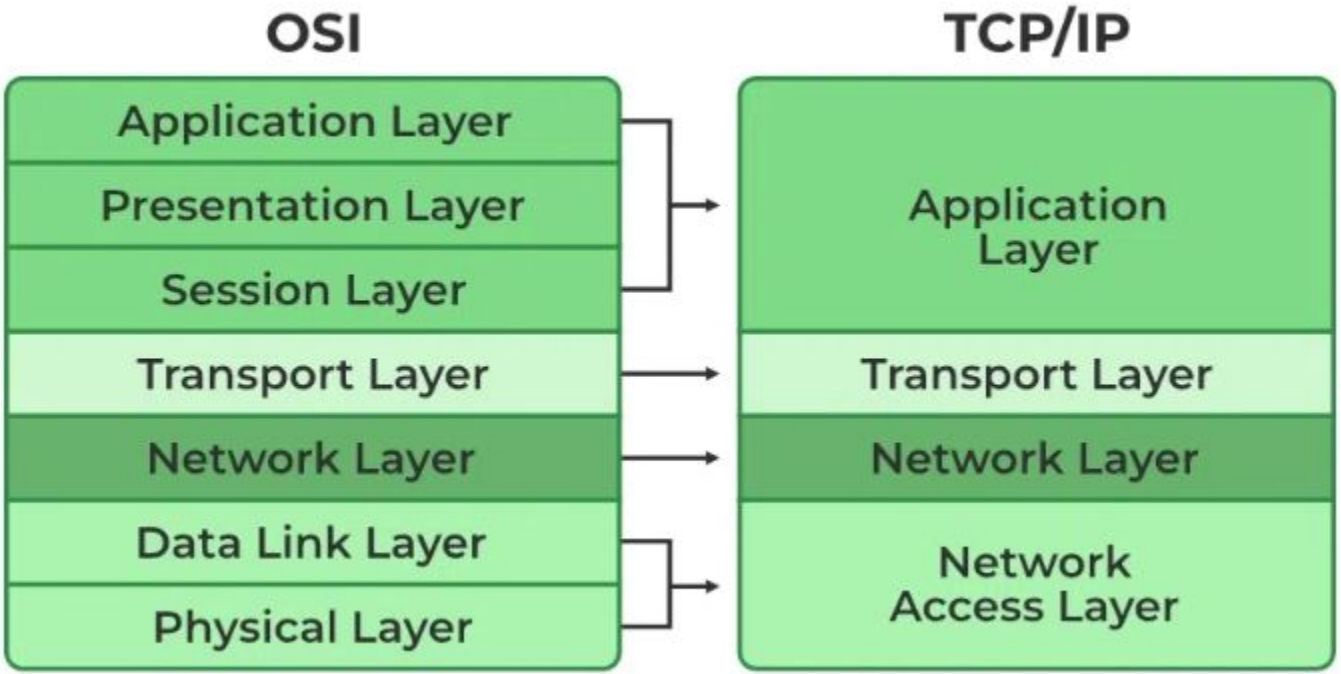
POP3 (Post Office Protocol) – Another protocol for email retrieval.

TCP/IP model

TCP/IP model

- **The TCP/IP model is a fundamental framework for computer networking.**
- **It stands for Transmission Control Protocol/Internet Protocol, which are the core protocols of the Internet.**
- **This model defines how data is transmitted over networks, ensuring reliable communication between devices.**
- **It consists of four layers: the Link Layer, the Internet Layer, the Transport Layer, and the Application Layer.**
- **Each layer has specific functions that help manage different aspects of network communication, making it essential for understanding and working with modern networks.**

TCP/IP and OSI model



Working of TCP\IP Model

- The main work of TCP/IP is to transfer the data of a computer from one device to another.
- The main condition of this process is to make data reliable and accurate so that the receiver will receive the same information which is sent by the sender.
- Whenever we want to send something over the internet using the TCP/IP Model, the TCP/IP Model divides the data into packets at the sender's end and the same packets have to be recombined at the receiver's end to form the same data, and this thing happens to maintain the accuracy of the data.
- TCP/IP model divides the data into a 4-layer procedure, where the data first go into this layer in one order and again in reverse order to get organized in the same way at the receiver's end.

Layers of TCP/IP Model

- Application Layer
- Transport Layer(TCP/UDP)
- Network/Internet Layer(IP)
- Network Access Layer

1. Network Access Layer

- The Network Access Layer represents a collection of applications that require network communication.
- This layer is responsible for generating data and initiating connection requests.
- It operates on behalf of the sender to manage data transmission, while the Network Access layer on the receiver's end processes and manages incoming data.
- The packet's network protocol type, in this case, TCP/IP, is identified by network access layer. Error prevention and "framing" are also provided by this layer.
- Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

2. Internet or Network Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows:

- **IP:**IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.
- **ICMP:**ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.
- **ARP:**ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP

2. Internet or Network Layer

The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet.

The Internet Layer is responsible for routing packets of data from one device to another across a network.

It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it.

Example: Imagine that you are using a computer to send an email to a friend. When you click “send,” the email is broken down into smaller packets of data, which are then sent to the Internet Layer for routing. The Internet Layer assigns an IP address to each packet and uses routing tables to determine the best route for the packet to take to reach its destination. The packet is then forwarded to the next hop on its route until it reaches its destination. When all of the packets have been delivered, your friend’s computer can reassemble them into the original email message.

In this example, the Internet Layer plays a crucial role in delivering the email from your computer to your friend’s computer. It uses IP addresses and routing tables to determine the best route for the packets to take, and it ensures that the packets are delivered to the correct destination. Without the Internet Layer, it would not be possible to send data across the Internet.

3. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

- **TCP:** Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by UDP , the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

4. Application Layer

The Application Layer in the TCP/IP model combines the functions of three layers from the **OSI model**: the **Application**, **Presentation**, and **Session** layers. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

The host-to-host layer is a layer in the OSI (Open Systems Interconnection) model that is responsible for providing communication between hosts (computers or other devices) on a network. It is also known as the transport layer.

Some common use cases for the host-to-host layer include:

- **Reliable Data Transfer:** The host-to-host layer ensures that data is transferred reliably between hosts by using techniques like error correction and flow control. For example, if a packet of data is lost during transmission, the host-to-host layer can request that the packet be retransmitted to ensure that all data is received correctly.
- **Segmentation and Reassembly:** The host-to-host layer is responsible for breaking up large blocks of data into smaller segments that can be transmitted over the network, and then reassembling the data at the destination. This allows data to be transmitted more efficiently and helps to avoid overloading the network.

- **Multiplexing and Demultiplexing:** The host-to-host layer is responsible for multiplexing data from multiple sources onto a single network connection, and then demultiplexing the data at the destination. This allows multiple devices to share the same network connection and helps to improve the utilization of the network.
- **End-to-End Communication:** The host-to-host layer provides a connection-oriented service that allows hosts to communicate with each other end-to-end, without the need for intermediate devices to be involved in the communication

The three main protocols present in this layer are:

- **HTTP and HTTPS:**HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.
- **SSH:**SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:**NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it’s out of sync.

TCP/IP	OSI
TCP refers to Transmission Control Protocol.	OSI refers to Open Systems Interconnection
TCP/IP uses both the session and presentation layer in the application layer itself.	OSI uses different session and presentatio layers.
TCP/IP follows connectionless a horizontal approach.	OSI follows a vertical approach.
The Transport layer in TCP/IP does not provide assurance delivery of packets.	In the OSI model, the transport laye provides assurance delivery of packets.
Protocols cannot be replaced easily in TCP/IP model.	While in the OSI model, Protocols are bette covered and are easy to replace with th technology change.

Difference between TCP/IP and OSI Model

TCP/IP model network layer only provides connectionless (IP) services. The transport layer (TCP) provides connections.	Connectionless and connection-oriented services are provided by the network layer in the OSI model.

Advantages of TCP/IP Mode

- **Interoperability** : The TCP/IP model allows different types of computers and networks to communicate with each other, promoting compatibility and cooperation among diverse systems.
- **Scalability** : TCP/IP is highly scalable, making it suitable for both small and large networks, from local area networks (LANs) to wide area networks (WANs) like the internet.
- **Standardization** : It is based on open standards and protocols, ensuring that different devices and software can work together without compatibility issues.
- **Flexibility** : The model supports various routing protocols, data types, and communication methods, making it adaptable to different networking needs.
- **Reliability** : TCP/IP includes error-checking and retransmission features that ensure reliable data transfer, even over long distances and through various network conditions.

Disadvantages of TCP/IP Model

- **Complex Configuration** : Setting up and managing a TCP/IP network can be complex, especially for large networks with many devices. This complexity can lead to configuration errors.
- **Security Concerns** : TCP/IP was not originally designed with security in mind. While there are now many security protocols available (such as SSL/TLS), they have been added on top of the basic TCP/IP model, which can lead to vulnerabilities.
- **Inefficiency for Small Networks** : For very small networks, the overhead and complexity of the TCP/IP model may be unnecessary and inefficient compared to simpler networking protocols.

- **Limited by Address Space** : Although IPv6 addresses this issue, the older IPv4 system has a limited address space, which can lead to issues with address exhaustion in larger networks.
- **Data Overhead** : TCP, the transport protocol, includes a significant amount of overhead to ensure reliable transmission. This can reduce efficiency, especially for small data packets or in networks where speed is crucial.
-

HTTP(Hypertext Transfer Protocol)

- The Hypertext Transfer Protocol (HTTP) is a protocol used mainly to access data on the World Wide Web.
- HTTP functions like a combination of FTP and SMTP.
- It is similar to FTP because it transfers files and uses the services of TCP.
- However, it is much simpler than FTP because it uses only one TCP connection.
- There is no separate control connection; only data are transferred between the client and the server.
- HTTP is like SMTP because the data transferred between the client and the server look like SMTP messages.
- HTTP uses the services of TCP on well-known port 80.

TELNET

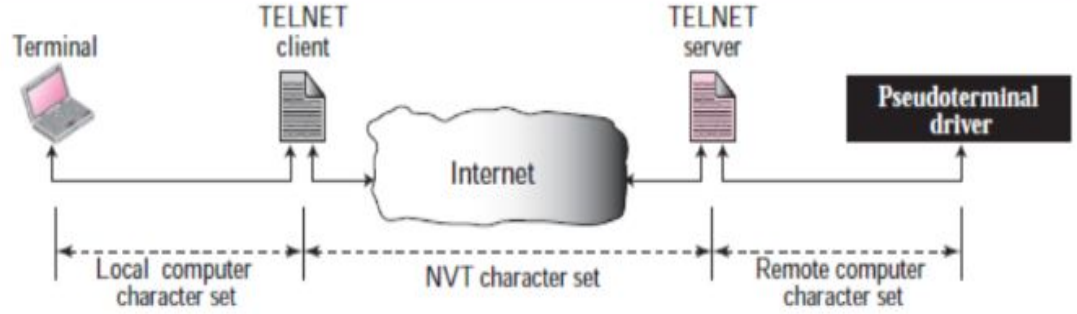
- It is an abbreviation for TErminaL NETwork.
- It is the standard TCP/IP protocol for virtual terminal service as proposed by ISO.
- It enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- The main task of the internet is to provide services to users.
- For example, users want to run different application programs at the remote site and transfers a result to the local site.
- This requires a client-server program such as FTP, SMTP. But this would not allow us to create a specific program for each demand.

- The better solution is to provide a general client-server program that lets the user access any application program on a remote computer.
- Therefore, a program that allows a user to log on to a remote computer. A popular client-server program Telnet is used to meet such demands.

Network Virtual Terminal (NVT)

- The mechanism to access a remote computer is complex. This is because every computer and its operating system accepts a special combination of characters as tokens.
- For example, the end-of-file token in a computer running the DOS operating system is Ctrl+z, while the UNIX operating system recognizes Ctrl+d.

Figure 20.3 *Concept of NVT*



IP (Internet Protocol):-

• IP is the host to host delivery protocol which belongs to the network layer & is designed for the Internet.

• The Internet Protocol (IP) is the connectionless transmission mechanism used by the TCP/IP protocol.

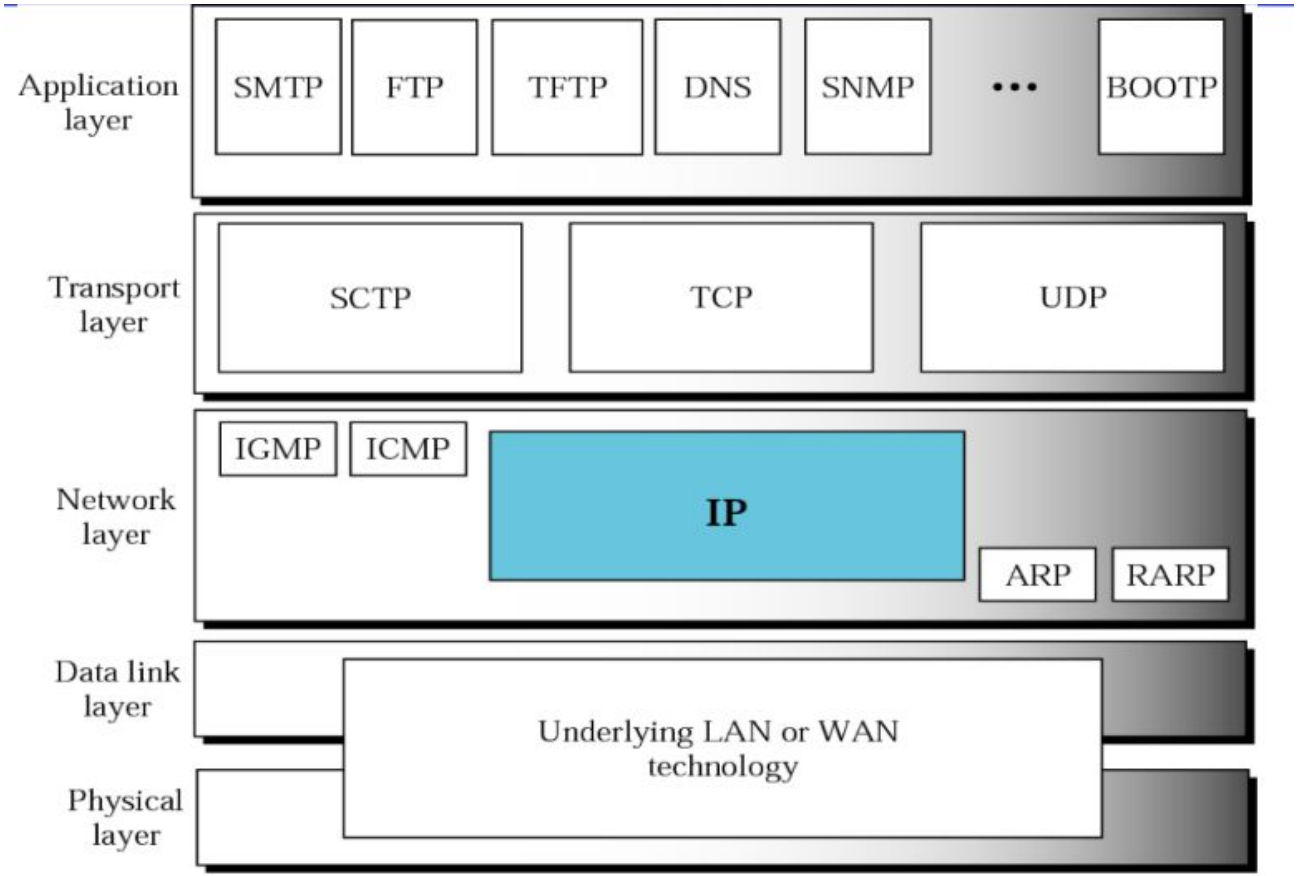
• IP is an unreliable and connectionless datagram—a best-effort delivery service. It ensures no guarantee of successfully transmission of data.

• The term best-effort means that IP provides no error checking or tracking.

• IP assumes the unreliability of the underlying layers and does its best to get a transmission through to its destination, but with no guarantees.

• If reliability is important, IP must be paired with a reliable protocol such as TCP

TCP / IP Protocol



UDP(User Datagram Protocol):

- Above Figure shows the relationship of the User Datagram Protocol (UDP) to the other protocols and layers of the TCP/IP protocol suite.
- UDP is located between the application layer and the IP layer, and serves as the intermediary between the application programs and the network operations.
- UDP is a connectionless, unreliable transport protocol.
- UDP provides a connectionless packet service that offers unreliable ‘best effort’ delivery. This means that arrival of packets is not guaranteed, nor is the correct sequencing of delivered packets.

- To create a process-to-process communication, UDP uses port numbers to accomplish this.
- To provide control mechanisms at the transport level, UDP does not provide flow control & acknowledgment for received packets.
- It does not add anything to the services of IP except for providing process-to-process communication.

Advantages of UDP

- UDP is a very simple protocol using a minimum of overhead.
- If a process wants to send a small message and does not care much about reliability, it can use UDP.
- Sending a small message using UDP takes much less interaction between the sender and receiver than using TCP.

Transmission Control Protocol (TCP)

- TCP is a reliable connection oriented protocol. i.e. a connection is established between the sender & receiver before the data can be transmitted.

- TCP divides the data it receives from the upper layer into segments & tags a sequence number to each segment which is used at the receiving end for reordering of data.

- TCP serves as the intermediary between application programs (application layer) & the network operations.(Network layer)

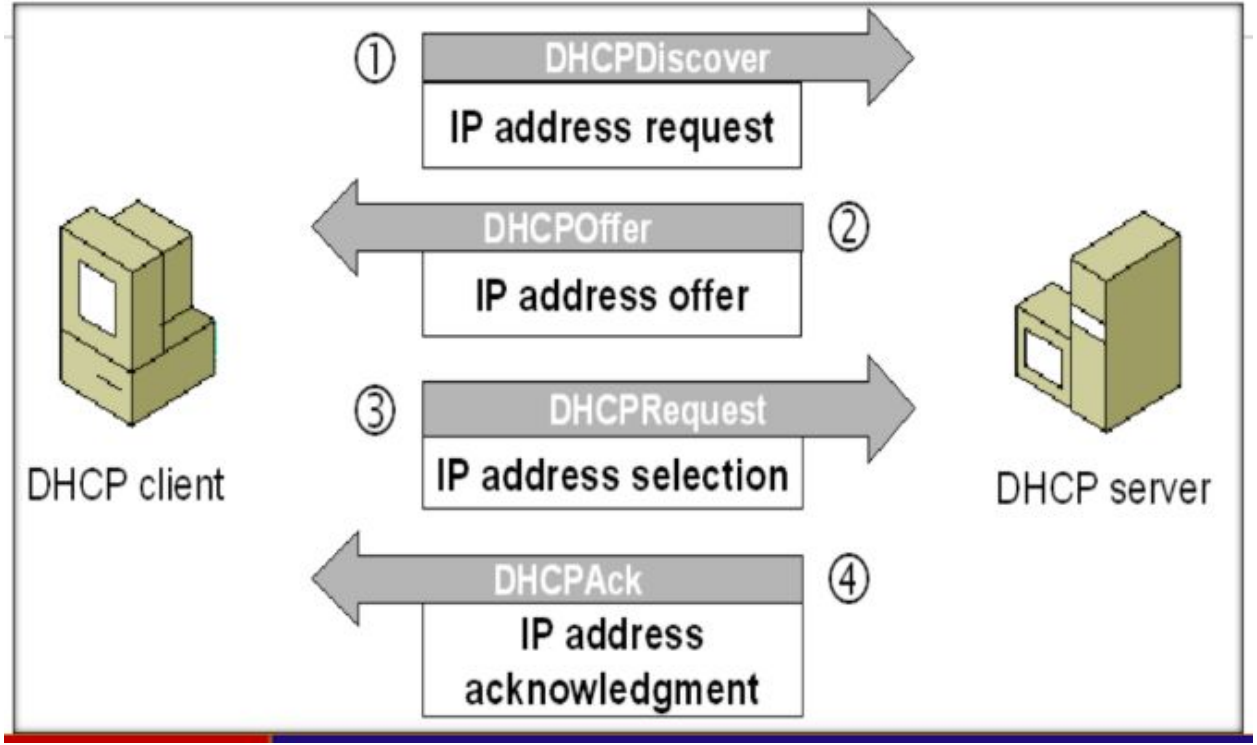
TCP Services

- Process-to-Process Communication
- Stream Delivery Service
- Full-Duplex Communication
- Multiplexing and Demultiplexing
- Connection-Oriented Service
- Reliable Service

Dynamic Host Configuration Protocol(DHCP)

- The Dynamic Host Configuration Protocol (DHCP) is a network management protocol used on UDP/IP networks whereby a DHCP server dynamically assigns an IP address.
- A DHCP server enables computers to request IP addresses and networking parameters automatically from the Internet service provider (ISP), reducing the need for a network administrator or a user to manually assign IP addresses to all network devices.
- In the absence of a DHCP server, a computer or other device on the network needs to be manually assigned an IP address, or to assign itself an APIPA address, which will not enable it to communicate outside its local subnet.
- Port number for DHCP is 67, 68.

DHCP Operation



DHCP OPERATION

1. DHCP discover message –

• This is a first message generated in the communication process between server and client. This message is generated by Client host in order to discover if there is any DHCP server/servers are present in a network or not. This message is broadcasted to all devices present in a network to find the DHCP server.

2. DHCP offer message –

• The server will respond to host in this message specifying the unleased IP address and other TCP configuration information. This message is broadcasted by server.

3. DHCP request message –

• When a client receives a offer message, it responds by broadcasting a DHCP request message.

4. DHCP acknowledgement message –

• In response to the request message received, the server will make an entry with specified client ID and bind the IP address offered with lease time.

Characteristics of DHCP

- Centralized and automated TCP/IP configuration.
- The efficient handling of IP address changes for clients that must be updated frequently, such as those for portable devices that move to different locations on a wireless network.
- The forwarding of initial DHCP messages by using a DHCP relay agent, which eliminates the need for a DHCP server on every subnet.
- The ability to define TCP/IP configurations from a central location.

Advantages of DHCP

- Easy use and configuration, as the network parameters have to be entered only once.
- Furthermore, the existing IP addresses can be used optimally.
- Because of the frequently changed locations, it is almost impossible that mobile devices to configure themselves constantly. Here, the automatic DHCP system offers the advantage of uncomplicated and simple applicability.

Disadvantages of DHCP

- The problem with DHCP is that clients accept any server. Accordingly, when another server is in the vicinity, the client may connect with this server, and this server may possibly send invalid data to the client.
- Security of MAC address is also not provided.

File Transfer Protocol(FTP)

FTP stands for File Transfer Protocol. It is an application layer protocol that moves the required files between local file systems and remote file systems. Like HTTP, it runs on top of TCP. To transfer a file, FTP uses two TCP connections in parallel:

- **Control connection (on port 21).**
- **Data connection (on port 20).**

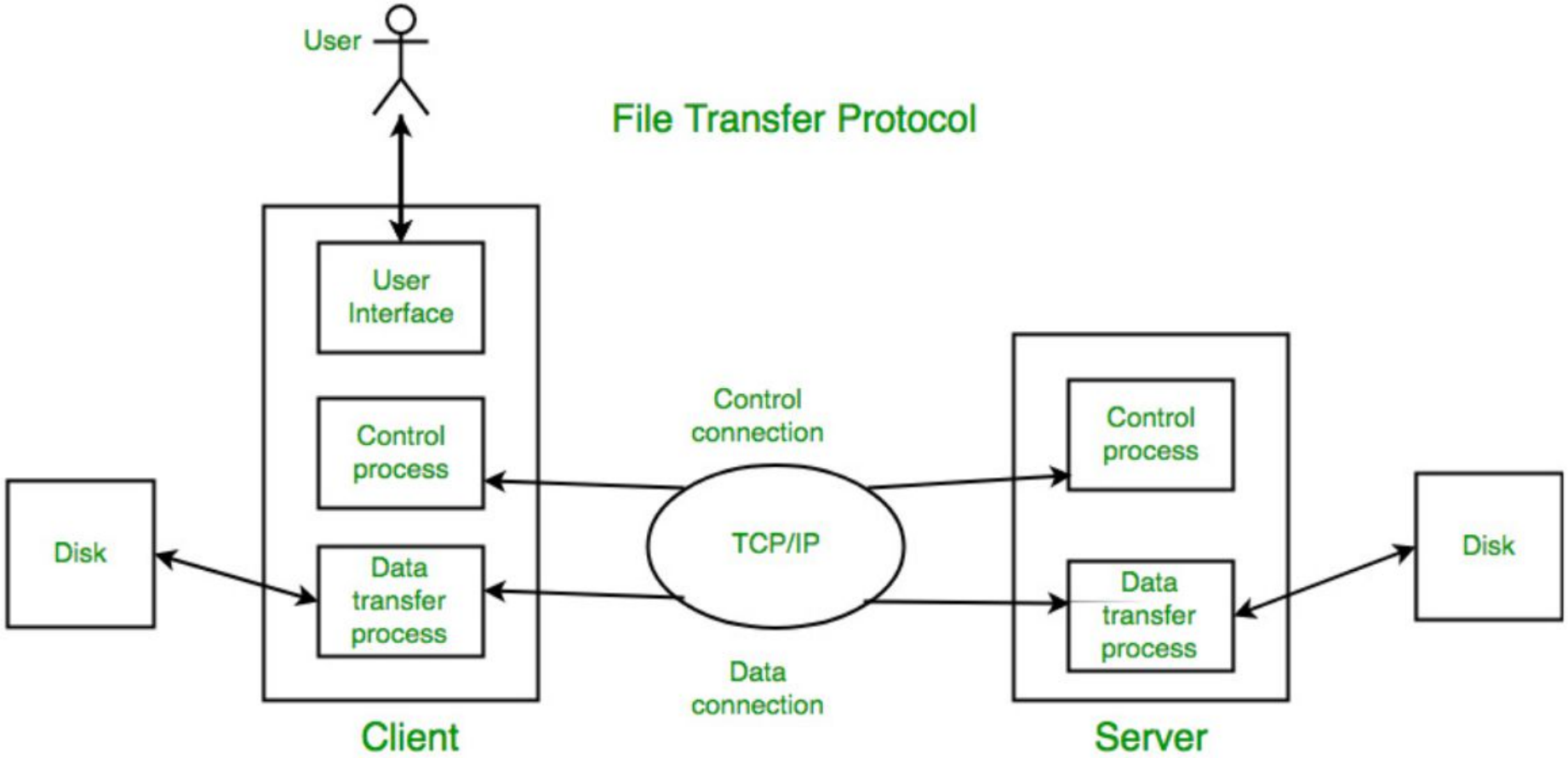
File Transfer Protocol(FTP)

FTP is a client server protocol that has two communication channel, command channel for conversation control and data channel for file content.

Here are steps mentioned in which FTP works:

- A user has to log in to FTP Server first, there may be some servers where you can access to content without login, known as anonymous FTP.
- Client can start a conversation with server, upon requesting to download a file.
- The user can start different functions like upload, delete, rename, copy files, etc. on server.

FTP can work on different modes like Active and Passive modes.


User
File Transfer Protocol
User Interface
Control process
Data transfer process
Disk
Client
Control connection
TCP/IP
Data connection
Control process
Data transfer process
Disk
Server

Types of Connection in FTP

- Control Connection
- Data Connection

Control Connection

For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of a control connection. The control connection is initiated on port number 21.

Data connection

For sending the actual file, FTP makes use of a data connection. A data connection is initiated on port number 20.

FTP sends the control information out-of-band as it uses a separate control connection. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.

FTP Session

When an FTP session is started between a client and a server, the client initiates a control TCP connection with the server side. The client sends control information over this. When the server receives this, it initiates a data connection to the client side. But the control connection remains active throughout the user session. As we know HTTP is stateless . But FTP needs to maintain a state about its user throughout the session.

FTP Clients

FTP works on a client-server model. The FTP client is a program that runs on the user's computer to enable the user to talk to and get files from remote computers. It is a set of commands that establishes the connection between two hosts, helps to transfer the files, and then closes the connection.

Features of FTP

- **It provides security to File Transfer.**
- **It gives its users well-maintained Access Control.**
- **It provides multiple Access Methods for File Transfer.**
- **It hides the details of an individual’s computer system.**

Advantages of FTP

- **FTP allows you to transfer multiple files along with multiple directories.**
- **FTP provides the ability to resume the interrupted file transfer.**
- **FTP gives the ability to add items to an upload/download queue.**
- **FTP is simple to use and implement.**
- **FTP is a connection-oriented protocol; Hence, it uses resilient control commands.**
- **FTP also allows backup facilities.**

Disadvantages of FTP

- **Lack of Security**
- **Vendor Equality is challenging to gain.**
- **FTP possesses Vulnerability to Attack.**
- **Monitoring of activity is quite difficult.**
- **FTP has a high latency rate because it’s a connection-oriented protocol.**

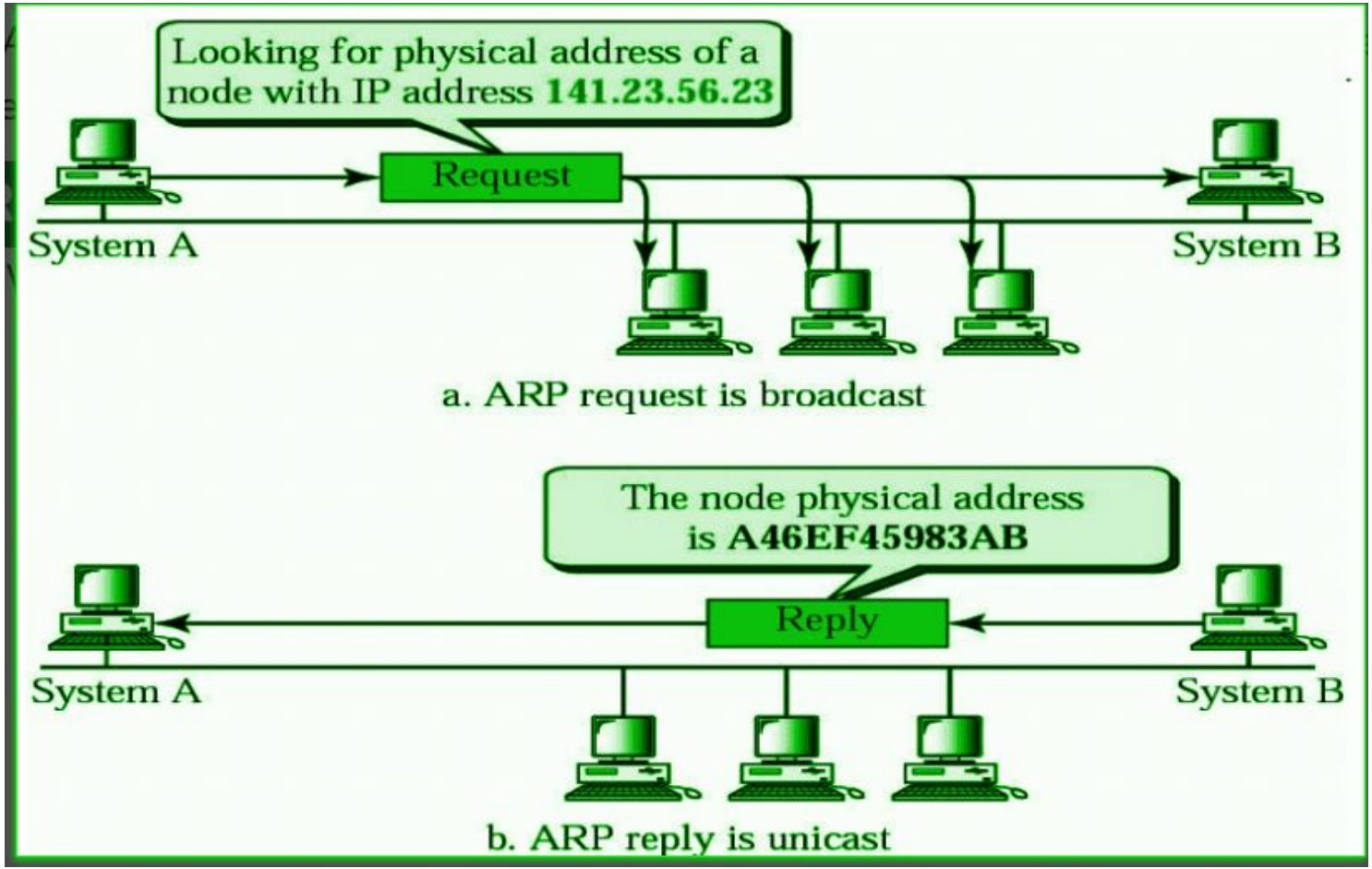
ARP Protocol (Address Resolution Protocol)

- It is an important protocol that plays an important role in the networking world.
- When working with your network systems, this protocol helps to identify specified network devices and find their addresses.
- Its main purpose is to duly transport data packets over the network, allowing them to move between devices connected to your heritage network.
- In other words, ARP is used to mapping the IP Address into MAC Address. When one device wants to communicate with another device in a LAN (local area network) network, the ARP protocol is used.

ARP Protocol (Address Resolution Protocol)

- **ARP protocol finds the MAC address based on IP address.**
- **IP address is used to communicate with any device at the application layer.**
- **But to communicate with a device at the data link layer or to send data to it, a MAC address is required.**
- **When data is sent to a local host, the data travels between networks via IP address.**
- **But to reach that host in LAN, it needs the MAC address of that host.**
- **In this situation the address resolution protocol plays an important role.**

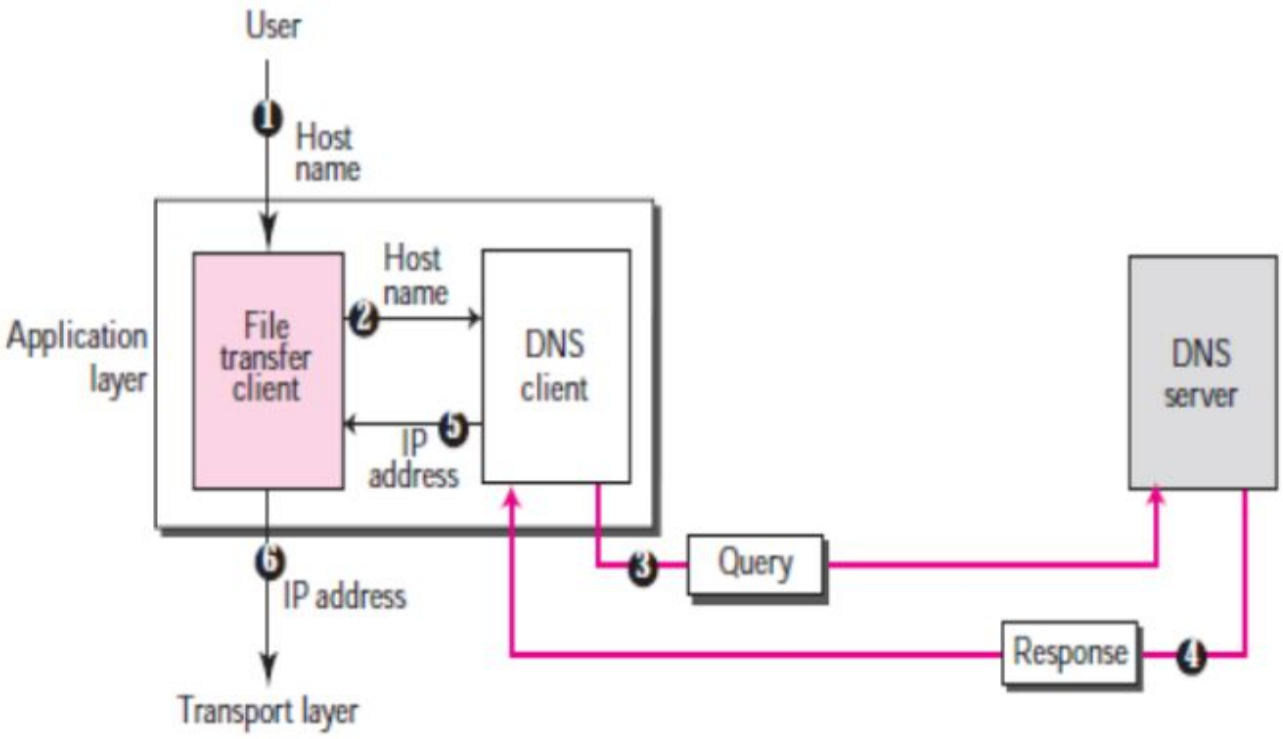
How ARP Protocol Works?



Working of address resolution protocol

- When a sender wants to communicate with a receiver, the sender first checks its ARP cache. Sender checks whether the receiver's MAC address is already present in the ARP cache or not?
- If the receiver's MAC address is already present in the ARP cache, the sender will communicate with the receiver using that MAC address.
- If the MAC address of the receiver device is not already present in the ARP cache, then in such a situation an ARP request message is prepared by the sender device.This message contains the MAC address of the sender, IP address of the sender and IP address of the receiver. The field containing the MAC address of the receiver is left blank because it is being searched.

DNS(Domain Name System)



DNS

- To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet. However, people prefer to use names instead of numeric addresses.
- Therefore, we need a system that can map a name to an address or an address to a name.

DNS

- **The names must be unique because the addresses are unique.**
- **A name space that maps each address to a unique name can be organized in two ways: flat or hierarchical.**

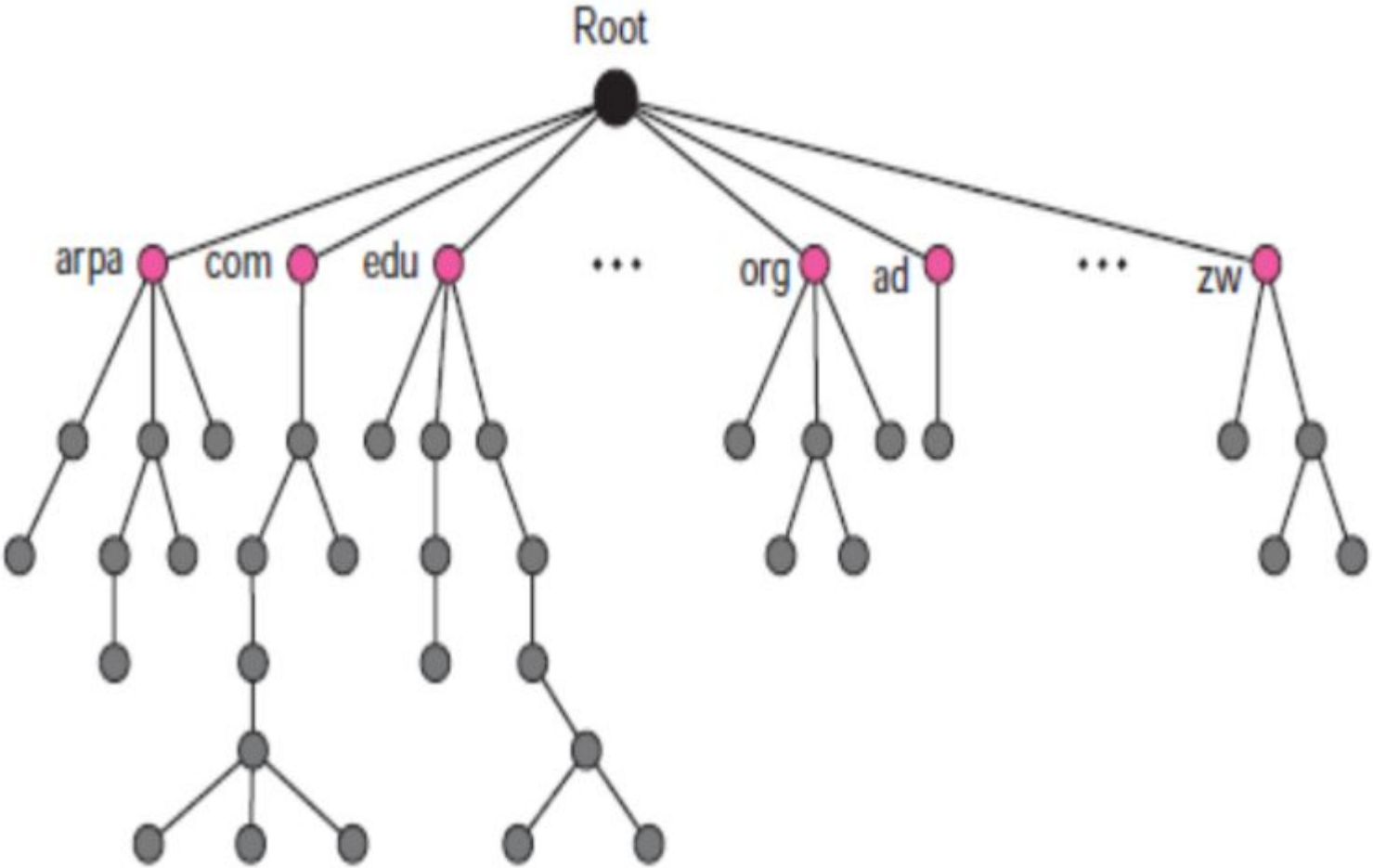
1. Flat Name Space

- **In a flat name space, a name is assigned to an address.**
- **A name in this space is a sequence of characters without structure.**
- **The names may or may not have a common section; if they do, it has no meaning.**
- **The main disadvantage of a flat name space is that it cannot be used in a large system such as the Internet because it must be centrally controlled to avoid ambiguity and duplication.**

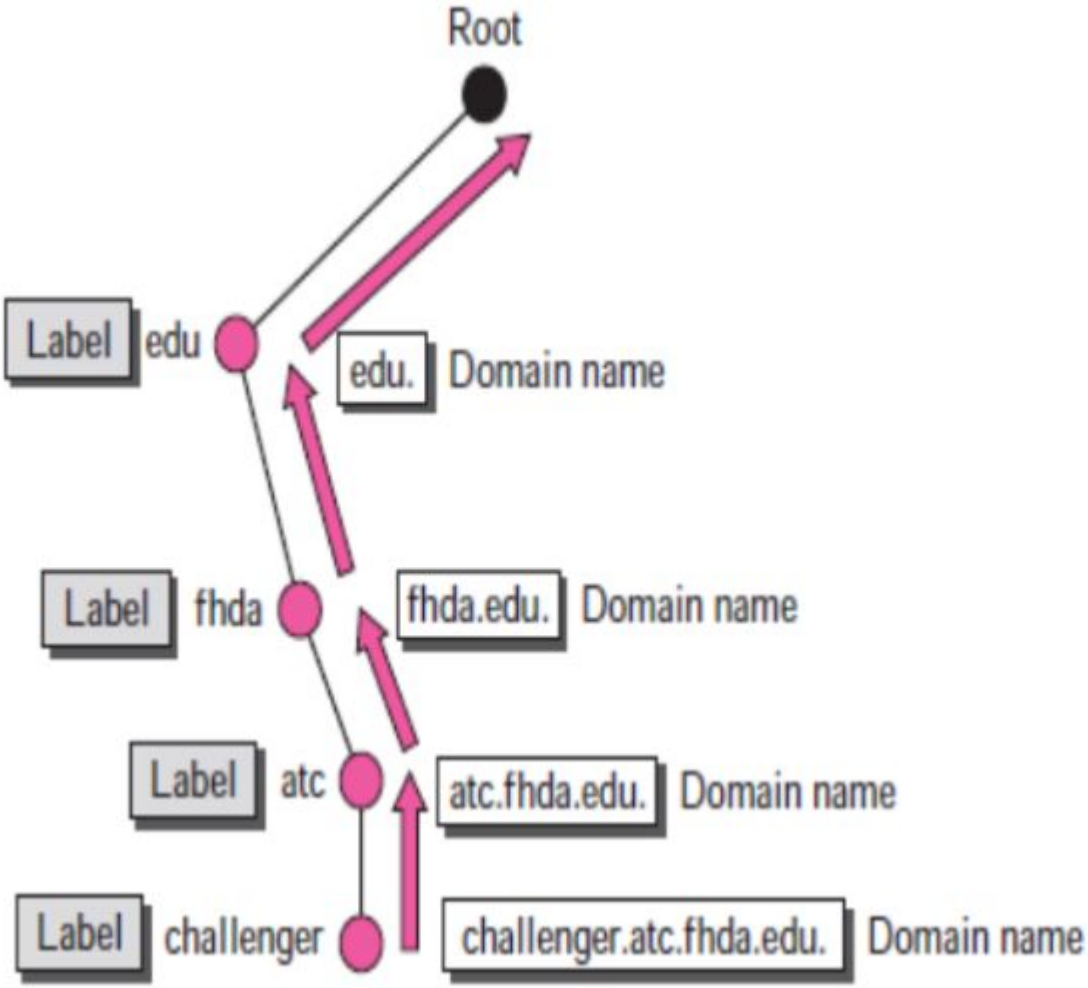
DNS

2. Hierarchical Name Space

- **In a hierarchical name space, each name is made of several parts.**
- **The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on.**
- **In this case, the authority to assign and control the name spaces can be decentralized.**


Root
arpa
com
edu
...
org
ad
...
zw

- **To have a hierarchical name space, a domain name space was designed.**
- **In this design the names are defined in an inverted-tree structure with the root at the top.**
- **The tree can have only 128 levels: level 0 (root) to level127**


Root
Label
edu
edu.
Domain name
Label
fhda
fhda.edu.
Domain name
Label
atc
atc.fhda.edu.
Domain name
Label
challenger
challenger.atc.fhda.edu.
Domain name

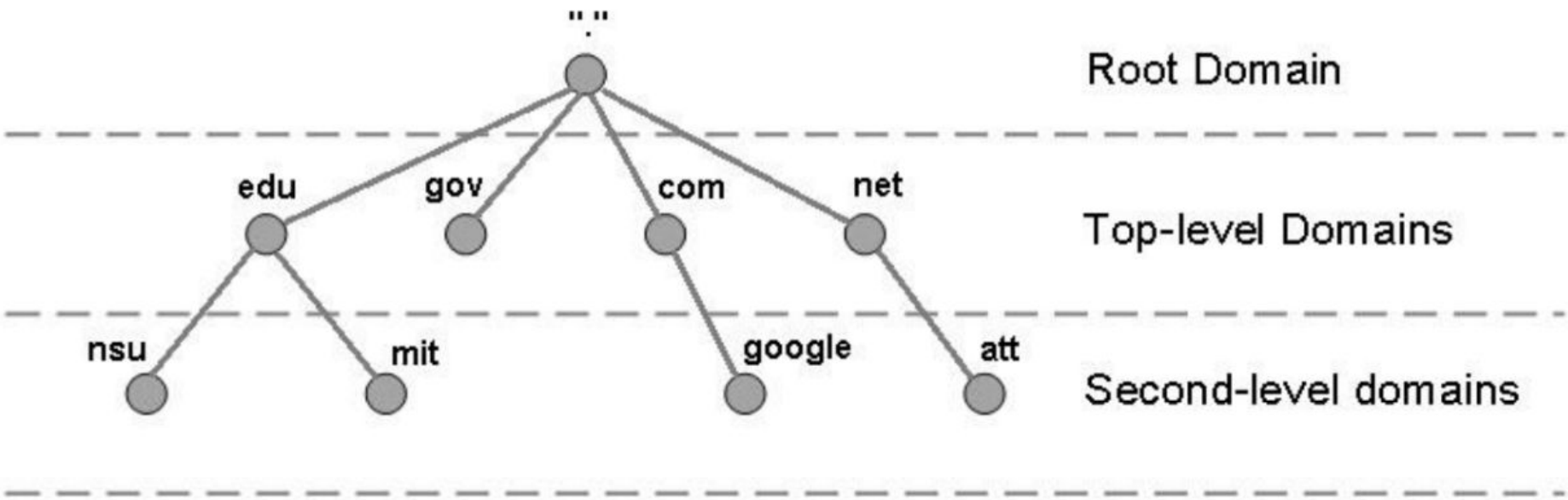
Label

- Each node in the tree has a label, which is a string with a maximum of 63 characters.
- The root label is a null string (empty string).
- DNS requires that children of a node (nodes that branch from the same node) have different labels, which guarantees the uniqueness of the domain names.

Domain Name

- Each node in the tree has a domain name.
- A full domain name is a sequence of labels separated by dots (.).
- The domain names are always read from the node up to the root.

Level of Domain Name Space



- Sender device broadcasts this ARP request message in the LAN. Because this is a broadcast message, every device connected to the LAN receives this message.
- All devices match the receiver IP address of this request message with their own IP address. Devices whose IP address does not match drop this request message.
- The device whose IP address matches the receiver IP address of this request message receives this message and prepares an ARP reply message. This is a unicast message which is sent only to the sender.

- In ARP reply message, the sender's IP address and MAC address are used to send the reply message. Besides, in this message the receiver also sends its IP address and MAC address.
- As soon as the sender device receives this ARP reply message, it updates its ARP cache with the new information (Receiver's MAC address). Now the MAC address of the receiver is present in the ARP cache of the sender. The sender can send and receive data without any problem.

Advantages of ARP Protocol

- By using this protocol we can easily find out the MAC Address of the device.
- There is no need to configure the end nodes at all to extract the MAC address through this protocol.
- Through this protocol we can easily translate IP address into MAC Address.
- There are four main types of this protocol. Which we can use in different ways, and they prove to be very helpful.

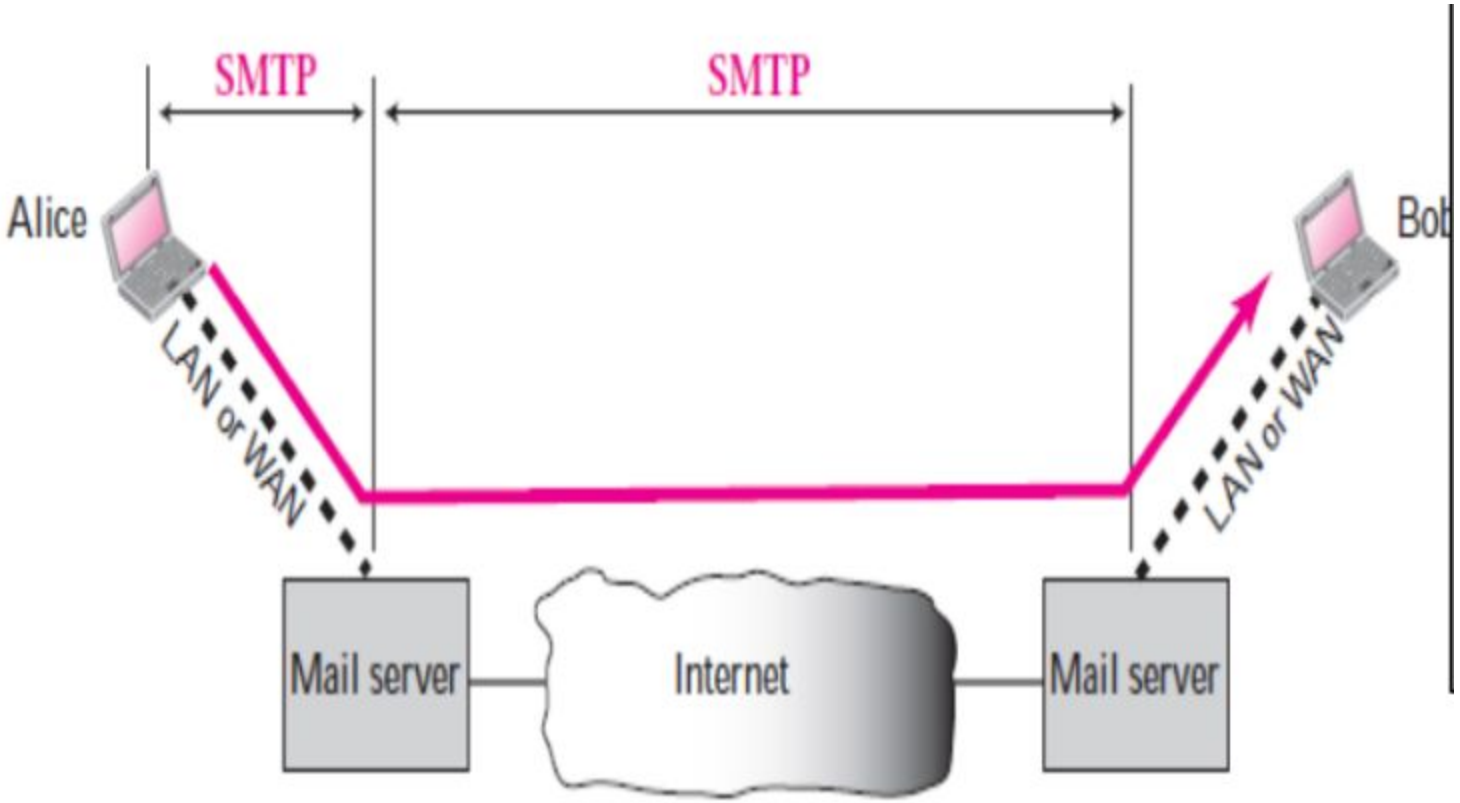
Simple Mail Transfer Protocol(SMTP)

- SMTP stands for Simple Mail Transfer Protocol.

• SMTP is a set of communication guidelines that allow software to transmit an electronic mail over the internet is called Simple Mail Transfer Protocol.

• It is a program used for sending messages to other computer users based on e-mail addresses.

• It provides a mail exchange between users on the same or different computers, and it also supports:


SMTP
SMTP
Alice
Bob
LAN or WAN
LAN or WAN
Mail server
Internet
Mail server

SMTP is used two times, between the sender and the sender’s mail server and between the two mail servers.

- Another protocol is needed between the mail server and the receiver.
- SMTP simply defines how commands and responses must be sent backand forth.
- Each network is free to choose asoftware package for implementation.